

FIN 620: Advanced Quantitative Financial Engineering

Term: Fall 2026 | Instructor: Dr. Fischer Black (fblack@quant.univ.edu)

Course Description & Overview:

Martingale pricing, stochastic volatility calibration, credit copulas, and high-frequency market microstructure execution.

COURSE ASSIGNMENTS & WEIGHT DISTRIBUTION

Assignment Description	Weight	Due Module	Deliverable Format
Stochastic Volatility & Heston Calibration	30%	Week 05	Python Volatility Surface Solver
Credit Risk & Copula Pricing Project	30%	Week 09	Monte Carlo Risk Engine
Algorithmic Execution System Capstone	30%	Week 12	Low-Latency Simulation Engine
Market Risk Quantitative Case Studies	10%	Continuous	Bi-Weekly Risk Briefs

WEEKLY TERM SCHEDULE & READING REQUIREMENTS

Timeline	Technical Topic Focus	Assigned Readings & Docs
Week 01	Module 01: Arbitrage Theory & Martingale Measures	Shreve (Ch. 1–3)
Week 02	Module 02: Brownian Motion and Ito Lemma	Shreve (Ch. 4 & 5)
Week 03	Module 03: Black-Scholes-Merton Partial DEs	Black & Scholes (1973)
Week 04	Module 04: Implied Volatility Surface & Smile	Gatheral (Ch. 1–3)
Week 05	Module 05: Local Volatility & Dupire Equation	Dupire (1994)
Week 06	Module 06: Stochastic Volatility: Heston Model	Heston (1993)
Week 07	Module 07: Interest Rate Term Structure: Vasicek	Hull & White (Ch. 28)
Week 08	Module 08: Heath-Jarrow-Morton (HJM) Models	Heath et al. (1992)
Week 09	Module 09: Credit Default Swaps & Default Risk	Merton (1974); ISDA Credit Specs
Week 10	Module 10: Monte Carlo Simulation & Variance	Glasserman (Ch. 2–4)
Week 11	Module 11: Optimal Execution & Microstructure	Almgren & Chriss (2000)
Week 12	Module 12: Trading System Evaluation & Review	Quantitative Finance Review